



S.A.P.D. Jain Pathashala

Walchand Institute of Technology, Solapur

An Autonomous Institute

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Department of Civil Engineering

Lab Details

Lab Name :	Geotechnical Engineering Lab
Lab Description	Major tests conducted in the laboratory 1. Index properties of soil 2. Shear strength and permeability of the soil 3. Bearing capacity determination of soil 4. Soil investigation and testing 5. Plate Load Test 6. Standard Penetration Test (SPT)
Lab Equipment	1. Liquid/plastic/shrinkage limit test apparatus 2. Direct shear test apparatus small size(60mmx60mm) (Motorised) 3. Direct shear test apparatus large size(300mmx300mm) (Motorised) 4. Standard and modified Proctor Compaction test apparatus 5. Permeability test apparatus 6. Unconfined compression test (Proving ring type) 7. Swell test apparatus 8. Vane shear test apparatus 9. Standard penetration test assembly 10. Consolidation test apparatus 11. Triaxial test apparatus (Motorized) 12. Automatic Compactor 13. Electric oven large size
Lab Floor Area in square meters	172
Practical engaged in Lab	1. Determination of Specific gravity of coarse- and fine-grained soil, 2. Particle size distribution – Mechanical sieve analysis, 3. Determination of Atterberg limits 4. Consistency limits- a. Liquid limit, b Plastic limit, c Shrinkage limit, 5. Determination of coefficient of Permeability,



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	6. Determination of Field density a) Core cutter b) Sand replacement method., 7. Direct box shear test, 8. Proctor compaction test- 9. Standard proctor, 10. Modified proctor, 8Triaxial test, 11. Unconfined compression test, 12. One-dimensional consolidation test, 13. Standard penetration test, 14. Plate load test, 15. Swelling pressure test, 16. Vane shear test
Name of Lab Incharge	Dr. Mahesh Girmallappa Kalyanshetti
Associated Faculty with the lab	1. Mr. Sandeep Bansilal Javheri 2. Mr. Avinash Angadi
Name of Lab Assistant	Mr. Shivaji V. Bhosale
Name/s of Menial Staff	Mr. Sunil V. Date